Going Public and Crowding Out Local Business Formation: Evidence from China*

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Abstract

This paper examines how local IPO activity affects business formation in China. Using a county-level panel spanning 1990–2022 and a staggered difference-in-differences design, we find that local business formation declines after a county experiences an IPO: new firm registrations and the stock of active firms fall by approximately 8%, while firm exit is not significantly affected. The decline in entry is concentrated among private firms and firms operating in the same industry as the IPO firm. Moreover, the effect is stronger following IPOs by firms with larger employment, but does not vary with either their capital or local access to finance, pointing to labor market reallocation as the primary mechanism. Consistent with this interpretation, the negative effect is more pronounced for state-owned enterprise IPOs and in less-developed regions, where competition for local labor is likely to be stronger. These findings show that IPOs can crowd out, rather than stimulate, local business formation, highlighting an important local resource reallocation effect of public listings.

Keywords: Initial public offerings; Business formation; Local spillovers; Crowding out; Real effects

JEL Codes: G14, G24, L26, R11

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1 Introduction

Initial public offerings (IPOs) are important milestones in firms' life cycles and are widely regarded as catalysts for firm growth (Borisov et al., 2021). Beyond their implications for listed firms, IPOs can generate broad spillover effects on the local economy, affecting local labor markets, innovation, the business environment, consumer spending, real estate markets, and migration, and are therefore often viewed as important drivers of regional economic development (Butler et al., 2019; Matray, 2021). Despite this growing literature, existing evidence from the United States remains mixed regarding whether IPOs stimulate local business formation in the surrounding local economy. On the one hand, a newly listed firm may generate information spillovers and reveal growth opportunities that encourage new business formation (Barrios et al., 2026). On the other hand, it may crowd out local business formation by absorbing scarce local resources such as skilled labor, financing, land, and government support (Cornaggia et al., 2024). Whether IPOs promote or discourage local business formation therefore remains an important but unresolved question.

China provides a markedly different institutional setting to answer this question. Local governments have strong incentives to cultivate and support firms seeking public listings through industrial and financial policies (Branstetter et al., 2023; Chen et al., 2018; Johansson et al., 2017). As a result, an IPO in China represents not only a corporate financing event but also the culmination of sustained regulatory review and local government support. This institutional feature may strengthen the resource reallocation effects associated with an IPO, as local resources and policy support may become increasingly concentrated on firms going public. At the same time, compared with developed economies, China also has a relatively large sector of micro and household businesses, many of which are established because of limited employment opportunities rather than the exploitation of entrepreneurial opportunities (Fang et al., 2023). Because these businesses rely heavily on the founders' own labor, improved local wage employment opportunities following an IPO may discourage potential entrepreneurs from entering self-employment, thereby amplifying the crowding-out effect on entrepreneurial entry.

A key empirical challenge is to identify the causal effect of IPOs on local

business formation. In China, occasional IPO suspensions expose even approved firms to uncertain listing delays and restrict access to public markets (Cong and Howell, 2021). More generally, both the eventual success and timing of an IPO are subject to substantial uncertainty. As a result, the timing of a county's first IPO reflects not only firm-level development but also external factors associated with the IPO approval process, making it plausibly independent of contemporaneous county-level economic conditions, including local business formation. From the perspective of the county, the timing of the first successful IPO therefore constitutes a plausibly exogenous shock. Exploiting this variation, we implement a staggered difference-in-differences design based on the year in which a county experiences its first successful IPO. Using a county-level panel spanning 1990–2022, we find that a county's first IPO reduces both new firm registrations and the stock of active firms by approximately 8 percent respectively, while having no significant effect on firm exit. These findings are robust to alternative estimators, model specifications, and sets of controls.

This paper further investigates the mechanisms underlying the crowding-out effect. We find that the negative spillovers are concentrated among firms operating in the same industry as the IPO firm, indicating that intensified competition plays an important role. To distinguish between competition for labor and financial resources, we examine how the effect varies with the characteristics of the IPO firm and local financial conditions. The crowding-out effect is significantly stronger for IPO firms with larger employment, but does not vary with local financial accessibility. Taken together, these findings point to labor market competition, rather than competition for financial resources, as the primary mechanism. Consistent with this interpretation, the negative effects are substantially stronger for state-owned IPO firms and in less-developed regions, where labor markets are likely to be less elastic and competition for labor and other local resources is likely to be more pronounced.

The labor market reallocation mechanism is also consistent with existing evidence from China. Prior research shows that employment creation plays an important role in the IPO approval process, as firms that generate excess employment are more likely to receive regulatory support, partly reflecting local governments' incentives to maintain employment (Johansson et al., 2017). Related

work finds that workers displaced by the closure of state-owned enterprises are more likely to enter entrepreneurship when wage employment opportunities deteriorate(Fang et al., 2023). Our findings suggest the opposite side of the same labor market adjustment: by creating new wage employment opportunities, an IPO may draw potential entrepreneurs into paid employment, thereby reducing entrepreneurial entry. Together, these findings suggest that changes in local wage employment opportunities can shift workers between paid employment and entrepreneurship, providing a natural labor market channel through which IPOs affect local business formation.

This paper makes several contributions to the literature.

First, this paper contributes to the emerging literature on the local spillover effects of IPOs. Existing studies have primarily focused on the consequences of IPOs for the firms that go public, documenting effects on myopic behavior, employment, productivity, innovation, and human capital (Babina et al., 2025; Bernstein, 2015; Borisov et al., 2021; Maksimovic et al., 2023). More recently, researchers have begun to examine how IPOs spill over to households, competitors, and the surrounding local economy. Butler et al. (2019) find that IPOs, as wealth and liquidity shocks, generate positive spillover effects on local labor markets, establishment growth, consumer spending, real estate, and migration. In the Chinese context, Lin et al. (2026) exploit unexpected IPO suspensions and show that local IPO activity reduces the profitability of nearby private SMEs. Other studies show that IPOs generate negative spillover effects on the stock prices and performance of competitors (HSU et al., 2010; Spiegel and Tookes, 2020), increase household stock market participation (Jiang et al., 2024), promote innovation among neighboring firms through local learning and labor mobility (Matray, 2021), and increase real estate prices (Hartman-Glaser et al., 2023). This paper extends this literature by examining how IPOs, as major capital market events, affect local business formation. To the best of our knowledge, this paper is the first to examine the effects of local IPOs on business formation in China and to document that IPOs crowd out local business formation. It further suggests that this crowding-out effect is primarily driven by labor market reallocation. More broadly, the findings highlight the importance of institutional context in shaping the local economic consequences of IPOs.

Within this literature, our paper is most closely related to Barrios et al. (2026) and Cornaggia et al. (2024), which provide contrasting evidence on the effects of IPOs on local business formation in the United States. Barrios et al. (2026) document positive effects of IPOs on business formation through information spillovers, while Cornaggia et al. (2024) find that IPOs that are large relative to the size of their counties reduce annual county-level establishment growth by 1.1 percentage points. We contribute to this literature in two main ways. First, we provide evidence from China, a large emerging economy with different institutional and labor market conditions, and document a substantial negative effect of local IPO activity: following an IPO, both new firm entry and the stock of active firms decline by approximately 8%. This contrasts with the positive effects found by Barrios et al. (2026) and points to a broader crowding-out effect than that documented by Cornaggia et al. (2024), whose negative effects are concentrated among relatively large IPOs. Second, we provide evidence on a labor market reallocation mechanism. We do not find stronger effects following IPOs by firms with larger capital, but we find substantially stronger crowding-out effects following IPOs by firms with larger employment. The effects are also stronger for state-owned enterprise IPOs and in less-developed regions, consistent with IPO firms drawing workers toward wage employment and away from entrepreneurship.

Second, this paper contributes to the literature on entrepreneurship and local business formation. A large body of research shows that business formation depends critically on the availability of finance and the allocation of capital across firms and projects. This literature emphasizes the roles of banking deregulation, credit market conditions, venture capital, and financial frictions in shaping new business creation and firm growth (Black and Strahan, 2002; Cetorelli and Strahan, 2006; Gompers et al., 2005; Kerr and Nanda, 2009; Michelacci and Silva, 2007; Samila and Sorenson, 2011). Another strand of the literature highlights the importance of politics, government institutions, and public policies in influencing business formation, showing that entry regulation (Barwick et al., 2025; Kaplan et al., 2011; Klapper et al., 2006), tax policies (Da Rin et al., 2011), and other political reasons (Akcigit et al., 2023; Chen et al., 2020) can substantially affect firm entry and business dynamism. Fang et al. (2023) also show that the large-scale closure

of state-owned enterprises in China during the late 1990s led to an increase in reluctant entrepreneurship. This paper extends this literature by showing that, in developing countries, major capital market events such as IPOs may discourage, rather than promote, business formation and firm entry. The findings suggest that, under resource constraints and imperfect institutions, the benefits of firms going public may come at the expense of local business formation.

Finally, this paper also contributes to the literature on how capital markets affect resource allocation and the real economy. A growing body of research shows that financial markets influence economic activity not only by relaxing financing constraints but also by reallocating scarce financial and productive resources across firms and sectors. Existing studies show that financial development and equity market liberalization promote investment and economic growth (Beck and Levine, 2004; Bekaert et al., 2005). Other studies document that sovereign credit ratings affect corporate investment by changing firms' financing conditions (Almeida et al., 2017), while government debt crowds out private-sector financing and investment through competition for financial resources (Demirci et al., 2019; Huang et al., 2020). More recently, research has shown that capital market frictions and institutions also shape real economic outcomes, such as bankruptcy and firm restructuring (Bernstein et al., 2019). This paper extends this literature by showing that IPOs, another important capital market event, can also reallocate local resources and thereby influence business formation. In particular, it shows that the local real effects of IPOs extend to business formation and depend importantly on institutional context.

The remainder of the paper proceeds as follows. Section 2 provides the institutional background and propose research hypothesis. Section 3 describes the data and introduces the empirical strategy. Section 4 presents the main results, robustness check, and heterogeneity analysis. Section 5 further discusses the mechanism underlying the main results. Section 6 concludes.

2 Institutional Background and Hypothesis

2.1 Institutional Background

Market-oriented Reforms and Business Registration in China. Since the initiation of market-oriented reforms in 1978, China has undergone a profound transition from a centrally planned economy toward a market-oriented economy. The gradual expansion of the private sector, decentralization of economic decision-making, and continuous improvement in market institutions have substantially stimulated business formation. Private firms have become an increasingly important source of economic growth, employment creation, and technological innovation. As shown in Figure A1, the number of new firm entrants increased rapidly over our sample period, while firm exits grew at a substantially slower pace. As a result, the stock of active firms expanded markedly over time. Our data shows that annual new registrations reached approximately 15 million by 2022, and the stock of active firms rose to around 90 million by 2022. At the same time, substantial regional differences in market development, government intervention, and factor-market distortions generated considerable heterogeneity in business formation across counties, providing a useful setting for studying how major economic events affect local business formation.

The rapid expansion of business formation has also been accompanied by continuous reforms to China's business registration system. Before the 2014 reform, establishing a new firm involved relatively cumbersome administrative procedures and minimum registered-capital requirements. The introduction of the registered-capital subscription system and the streamlining of administrative approval procedures substantially lowered entry barriers and reduced the cost of starting a business, contributing to a further surge in new business registrations. These reforms also improved the coverage and quality of administrative registration records. Because formal business operations generally require registration, these data provide a comprehensive measure of firm creation and have been widely used to study business formation in China.

China's IPO System has undergone several rounds of institutional reform since the establishment of the Shanghai and Shenzhen Stock Exchanges in 1990

and 1991. During the early stage of capital market development, IPOs were governed by an administrative approval system with issuance quotas, under which regulators exercised direct control over both the number of listed firms and the allocation of listing opportunities. Beginning in the early 2000s, the quota system was gradually replaced by a merit-based approval regime that strengthened disclosure requirements and standardized listing criteria while retaining substantive regulatory review. More recently, China initiated a gradual transition toward a registration-based IPO system, starting with the STAR Market in 2019 and later expanding to other boards before the full implementation of registration-based IPOs in 2023.

Our analysis focuses exclusively on firms listed on China's A-share market (the Shanghai and Shenzhen Stock Exchanges). China piloted a registration-based IPO system on the STAR Market in 2019, but did not extend the reform to the broader A-share market until 2023, after our sample period (Li et al., 2026; Qian et al., 2024). Therefore, the IPOs examined in our study (1990–2022) were conducted under the traditional approval-based regulatory framework, in which listing decisions were subject to substantive review by the China Securities Regulatory Commission (CSRC).

Unlike market-oriented IPO systems, where firms generally have substantial discretion over the timing of going public once regulatory requirements are met, China's IPO market has historically been characterized by significant administrative intervention. In addition to reviewing firms' eligibility, the CSRC exercised considerable control over the pace of IPO approvals. The regulator periodically adjusted the number of approved IPOs, slowed or accelerated the review process, and occasionally suspended new IPO approvals altogether in response to macroeconomic conditions, market performance, or broader policy objectives. As a result, firms that had already completed the application process often faced uncertain waiting periods before receiving approval, and otherwise comparable firms could complete their IPOs at substantially different times because of regulatory scheduling rather than differences in firm quality or local economic conditions.

This institutional setting has important implications for our identification strategy. Although firms self-select into the IPO application process, the exact

timing of IPO completion is not determined solely by firms' own decisions or local market conditions. Instead, it is jointly shaped by regulatory review, administrative capacity, and issuance scheduling. Consequently, conditional on firms entering the IPO approval process, the timing of IPO events can be viewed as plausibly exogenous to contemporaneous entrepreneurial activities within a county, providing a useful source of variation for identifying the local spillover effects of IPOs.

2.2 Theoretical Framework and Hypotheses

The effect of a local IPO on business formation is theoretically ambiguous. On the one hand, going public may stimulate local business formation by improving information transparency, increasing investor attention, and enhancing the visibility of local industries. The listing of a local firm may also generate knowledge spillovers, expand upstream and downstream business opportunities, and strengthen local entrepreneurial ecosystems (e.g., Barrios et al., 2026).

On the other hand, a newly listed firm typically experiences substantial expansion following the IPO, resulting in increased demand for labor, capital, land, managerial talent, and business services. In regions where these production factors are imperfectly elastic, competition for scarce resources may increase the cost of starting new businesses and discourage entrepreneurial entry. This mechanism may be particularly important in China, where local governments often devote considerable administrative attention and policy support to listed firms, potentially reallocating public resources toward newly listed firms and away from prospective entrepreneurs. Consequently, a local IPO may crowd out business formation through the reallocation of scarce local resources.

These competing mechanisms can be illustrated by a simple entry model. Suppose that a potential entrepreneur enters the market if the expected profit from establishing a firm is nonnegative:

$$\pi_{ict} = A_{ct} - w_{ct} - r_{ct} - \phi_{ict},$$

where A_{ct} denotes local business opportunities, w_{ct} represents labor costs, r_{ct} captures the cost of finance and other local production inputs, and ϕ_{ict} is an entrepreneur-specific entry cost. Following the standard entry literature (e.g.,

Hopenhayn, 1992), entrepreneurial entry occurs whenever $\pi_{ict} \geq 0$.

A local IPO affects entrepreneurial entry through two opposing channels. First, the IPO may increase A_{ct} by improving information transparency, generating knowledge spillovers, expanding business opportunities, and strengthening local production networks. Second, the IPO may increase w_{ct} and r_{ct} by intensifying competition for labor, financing, land, managerial talent, and government-supported resources. Consequently, the overall effect of an IPO depends on whether the positive information spillovers outweigh the negative resource reallocation effects.

If the information channel dominates, business formation should increase following a local IPO.

Hypothesis 1. *A local IPO increases business formation by improving information transparency, enhancing industry visibility, and expanding local business opportunities.*

Alternatively, if competition for scarce local resources dominates, business formation should decline following a local IPO.

Hypothesis 2. *A local IPO reduces business formation by reallocating scarce local resources toward the newly listed firm, thereby crowding out potential entrepreneurs.*

Hypothesis 2 yields several empirical predictions. First, if resource competition is an important mechanism, the crowding-out effect should be stronger among firms operating in the same industry as the IPO firm, since they compete more directly for both production factors and product market opportunities. If labor market competition is the primary channel, the crowding-out effect should increase with the employment size of the IPO firm but not with its capital size. By contrast, if financial competition is an important mechanism, the crowding-out effect should be stronger in counties with lower financial accessibility. Finally, because state-owned enterprises typically receive greater government support and because production factors are generally more scarce in less-developed regions, the crowding-out effect should also be more pronounced when the IPO firm is state-owned and in less-developed regions.

3 Empirical Strategy

3.1 Data

The empirical analysis uses a county-year panel covering the period from 1990 to 2022. The core panel is constructed from China's National Enterprise Registration System, the official nationwide business registration and disclosure system administered by the market regulatory authorities. Under Chinese law, all business entities are required to register in this system, making it the most comprehensive source of firm-level registration information in China. Using these records, we aggregate firm-level information to the county-year level and construct three measures of local business formation: the annual number of newly registered firms, the stock of active firms at the end of each year, and the annual number of firm exits. These county-level measures are then merged with IPO records, which identify the first year in which a county hosts a listed firm, and with county socioeconomic characteristics compiled from official county statistical sources.

We assign each IPO firm to a county based on the firm's registered address at the time of listing. A county is therefore considered to experience an IPO when a firm registered in that county goes public on China's A-share market. Although a firm's registered address may differ from its office location or actual place of operation, the registered location remains economically relevant because firms are subject to important administrative oversight by the local government at the place of registration and maintain fiscal ties with that jurisdiction, including through local taxation. We therefore use the registered county as the relevant local market for assigning IPO events and examining their effects on local business formation.

3.2 Variables

Dependent Variables. The dependent variables are constructed from the county-level business registration data aggregated from China's National Enterprise Registration System. Specifically, let $Entry_{ct}$ denote the number of newly registered firms in county c during year t, $Stock_{ct}$ denote the number of active firms at the end of year t, and $Exit_{ct}$ denote the number of firm exits during year t. These three variables capture the flow of entrepreneurial entry, the stock of active firms, and business closures, respectively. The estimating equations use the

logarithmic transformations $\log(Entry_{ct} + 1)$, $\log(Stock_{ct} + 1)$, and $\log(Exit_{ct} + 1)$. The log-plus-one transformation accommodates county-year observations with zero.

Control Variables. County-level statistical yearbooks are only consistently available from 2000 onward. Consequently, constructing contemporaneous annual county-level controls would substantially shorten the sample period by excluding much of the pre-2000 period and weaken the comparability between treated and control counties over the long horizon covered by our analysis. To address this limitation, we adopt a interaction-control strategy following established practice in the literature, such as BIERMANN and HUBER (2024) and Martin (2025). Specifically, we construct the control block using county characteristics measured in 2010 and interact these baseline characteristics with year fixed effects. This specification allows counties with different economic conditions and levels of development to follow flexible time paths, while avoiding the substantial sample truncation that would arise from using annually varying controls, which are only available from 2000 onward. The baseline controls include local public expenditure, the number of above-scale industrial firms, county GDP, county population, and the value added of the secondary and tertiary industries. We use their 2010 values because county-level data are most complete and consistently available in that year. In the regressions, the logarithms of public expenditure, industrial firms, GDP, and population are used, while the secondary- and tertiary-industry variables enter in levels.

3.3 Summary Statistics

We first present summary statistics in Table 1 for the entry, exit, and stock of different types of firms in the county-year panel. We also report summary statistics for the treatment indicator, Post-IPO. This indicates that, on average, approximately 2,100 firms are newly registered and 350 firms exit in each county-year, while the average stock of active firms is around 11,000 per county. The number of firm exits is likely to be substantially underestimated because many firms cease operations without formally deregistering and instead remain in the registry as inactive or "zombie" firms. In terms of the distribution across firm types, private firms account

for more than 98% of firm entry, exit, and stock. Private firms are established by individual entrepreneurs and include small, micro, and household businesses. They are numerous but typically very small in scale, making them a close proxy for local business formation.

In Table 1, the IPO treatment indicator shows that approximately 20% of county-year observations are exposed to the IPO shock and therefore classified as treated. Turning to the timing of counties' first IPOs, a substantial share of treatment events occurred before 2000 as shown in Figure 1. In addition, we report summary statistics for county characteristics. Table A1 reports the mean values of the 2010 control variables for the full sample, treated counties, and control counties. Treated and control counties differ substantially in their levels of economic development, particularly in the number of above-scale industrial firms, industrial structure, and total GDP.

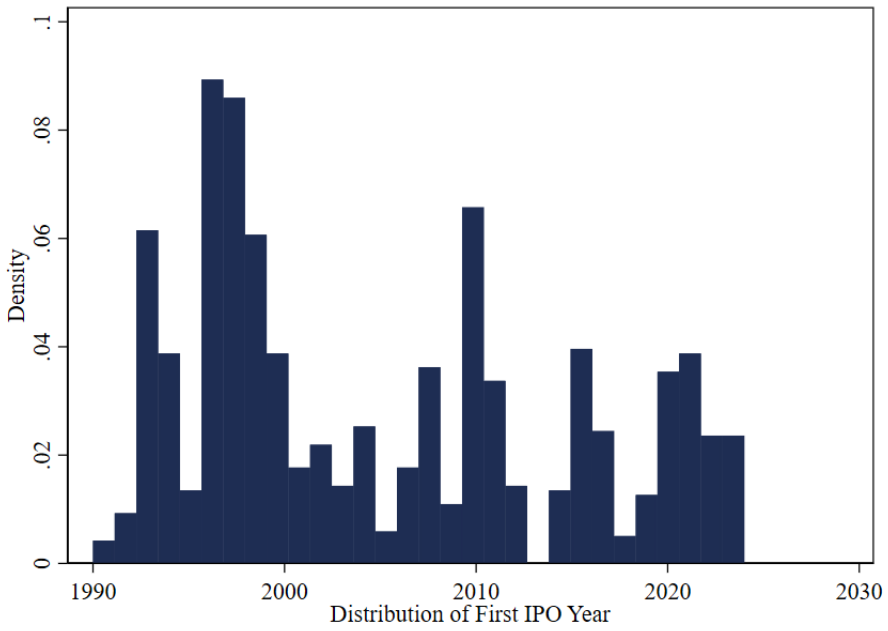


Figure 1: Distribution of County's First IPO Year

Notes: This figure shows the distribution of the year in which each treated county experiences its first IPO in our sample.

3.4 Identification

To identify the causal effect of a local IPO on business formation, we employ a staggered difference-in-differences (DiD) design similar to that of Barrios et al.

Table 1: Summary Statistics: County-Year Level

Variable	Obs	Mean	Std. dev.	Min	Max
Firm entry					
All firms	52,746	2,127.17	6,432.08	1	442,070
SOEs	52,746	8.19	25.41	0	1,017
Private firms	52,746	2,099.64	6,424.54	0	441,654
Foreign firms	52,746	19.34	52.92	0	1,815
Firm stock					
All firms	52,746	11,181.22	27,500.66	0	1,169,186
SOEs	52,746	46.02	184.96	0	6,770
Private firms	52,746	10,975.57	27,393.42	0	1,164,323
Foreign firms	52,746	159.63	291.86	0	4,215
Firm exit					
All firms	52,746	346.43	1,027.61	0	29,982
SOEs	52,746	3.88	12.58	0	383
Private firms	52,746	326.35	1,019.66	0	29,965
Foreign firms	52,746	16.20	50.47	0	2,077
Post IPO	52,746	0.192	0.394	0	1
Cumulative IPOs	52,755	.495	1.852	0	41

Notes: This table reports summary statistics for the main variables used in the analysis. Firm entry is the number of newly registered firms in a county-year, firm stock is the number of active firms, and firm exit is the number of firms exiting in a county-year. SOEs include state-owned enterprises, collective enterprises, and public institutions; private firms include private enterprises and self-employed businesses; and foreign firms include foreign-invested enterprises. Post IPO is an indicator equal to one for county-years following a county's first IPO and zero otherwise. The cumulative IPOs measure captures the total number of firms that have gone public in a county up to each year.

(2026). We classify counties that experience at least one IPO during the sample period as treated counties, while counties that never experience an IPO serve as the control group. An IPO is assigned to a county based on the registered address of the firm at the time of listing. For treated counties, treatment timing is defined as the year of the county's first IPO. The empirical analysis therefore compares changes in entrepreneurial outcomes before and after the first IPO in treated counties with contemporaneous changes in counties that never host an IPO. Although some treated counties experience multiple IPOs during the sample period, approximately 60% experience only one or two IPOs, implying that the first IPO captures the primary treatment for most treated counties. The baseline specification therefore uses the first IPO to define treatment timing.

In the robustness analysis in 4.3, we address the presence of multiple IPOs in two ways. First, we restrict the sample to counties with only one or two IPOs to assess whether the baseline estimates are driven by counties experiencing repeated listing events. Second, we replace the first-IPO treatment indicator with the cumulative number of IPO firms in each county and year, thereby using all IPO events in the sample and estimating the marginal effect of an additional local IPO on business formation.

The identification strategy exploits the institutional features of China's IPO approval process. Although firms choose whether and when to initiate the listing process, the ultimate completion and timing of an IPO also depend on regulatory approval and other external factors beyond firms' direct control, leaving the process subject to substantial policy uncertainty (Cong and Howell, 2021). Consequently, the occurrence of a county's first IPO is not solely determined by local economic conditions or firms' own decisions, providing a plausibly exogenous source of variation for identifying the local effects of IPOs. The identification further relies on the standard parallel trends assumption that, absent an IPO, business formation would have evolved similarly in treated and control counties. Although the parallel trends assumption itself is not directly testable, one of its testable implications is that treated and control counties exhibit similar pre-treatment trends, rather than identical levels, of business formation.

To further isolate the local effects of IPOs, all specifications include county fixed effects and year fixed effects. More demanding specifications additionally include

province-by-year fixed effects, so that identification comes from comparisons between counties within the same province that experience their first IPO at different times or never experience an IPO during the sample period. This absorbs province-specific macroeconomic conditions, policy changes, and common regional shocks that could otherwise confound the estimated effects. The preferred specifications further allow for differential trends based on county characteristics.

3.5 Model Specification

This paper employs a staggered difference-in-differences design. The baseline specification is

$$Y_{ct} = \beta IPO_{ct} + \eta_c + \lambda_t + \gamma_{pt} + X_c' \lambda_t + \varepsilon_{ct}, \tag{1}$$

where Y_{ct} denotes outcome variables for county c in year t. The core explanatory variable is the county's first-IPO treatment indicator. Let T_c be the first year in which county c hosts a local IPO. The baseline treatment indicator IPO_{ct} equals one if $t \geq T_c$ for treated counties and zero otherwise. It therefore captures the post-treatment period after a county first enters the public equity market through a locally headquartered listed firm.

In addition, η_c denotes county fixed effects, λ_t denotes year fixed effects, and γ_{pt} denotes province-by-year fixed effects. The term $X_c' \lambda_t$ captures interactions between county characteristics and year fixed effects. In practice, X_c includes the 2010 values of the logarithms of local public expenditure, the number of above-scale industrial firms, county GDP, and county population, together with the shares of secondary and tertiary industries in county GDP. The coefficient of interest, β, measures the average change in business formation outcomes after the first local IPO relative to counties that have not yet been treated or are never treated, conditional on the fixed effects and trend controls. The identifying assumption is that, absent treatment, treated and comparison counties would have followed parallel trends after conditioning on these controls. Standard errors are clustered at the county level.

The validity of the difference-in-differences design relies on the parallel trends assumption, which requires that, in the absence of treatment, outcomes in the treated and control groups would have evolved similarly over time. Under this assumption, the untreated counties provide a credible counterfactual for estimating

the Average Treatment Effect on the Treated (ATT). Although the counterfactual outcome cannot be observed directly and the assumption is therefore not testable, its plausibility can be evaluated by examining whether treated and control units display comparable trends prior to treatment. To this end, we estimate an event-study specification that allows us to assess pre-treatment dynamics and trace the evolution of treatment effects following the IPO. Specifically, we estimate the following equation:

$$Y_{ct} = \sum_{k=-7, k\neq -1}^{7} \beta_k \mathbf{1}\{t - T_c = k\} + \eta_c + \lambda_t + \gamma_{pt} + X_c' \lambda_t + \varepsilon_{ct}, \tag{2}$$

where the omitted category is $k = -1$. The event-study coefficients trace the timing of effects before and after a county's first IPO and provide a direct visual assessment of whether treated and untreated counties exhibit comparable pre-treatment dynamics.

4 Empirical Results

4.1 Baseline Results

This section presents the baseline estimates of the effects of a county's first IPO on new firm entry, firm stock, and firm exit using a staggered difference-in-differences (DiD) design. Table 2 reports the effect of a county's first IPO on new firm entry. Column (1) includes county and year fixed effects only and yields a large negative coefficient of -0.421. As the specification becomes more demanding, the magnitude declines, which is expected once broader regional shocks and county-specific trends are absorbed. The preferred specification is column (4), which adds province-by-year fixed effects together with interactions between county characteristics and time fixed effects. The estimated coefficient remains negative and statistically significant at the 5% level, with a value of -0.084. Interpreted semi-elastically, this implies that the first local IPO is associated with approximately a 8.4% decline in new firm registrations.

Table 3 presents the corresponding results for the stock of active firms. The estimates follow a similar pattern across specifications. In the preferred specification, the coefficient is -0.080 and statistically significant at the 5% level,

implying an 8% reduction in the stock of incumbent firms. Table 4 examines whether this decline in firm stock is driven by an increase in firm exit. Across all specifications, the estimated coefficients are small and statistically insignificant. The comparable declines in firm entry and firm stock, together with the absence of a significant effect on firm exit, suggest that a county's first IPO primarily reduces the formation of new firms rather than accelerating the closure of existing ones. This pattern is consistent with a decline in the local supply of potential entrepreneurs, which gradually translates into a persistent reduction in the stock of active firms.

The results lend greater support to Hypothesis 2 than to Hypothesis 1, suggesting that the negative resource reallocation effects of IPOs outweigh their positive information spillovers and business opportunity effects on local business formation. Our findings are more consistent with Cornaggia et al. (2024), who show that large IPOs reduce county-level establishment growth, than with Barrios et al. (2026), who document positive entrepreneurial spillovers from IPOs. We therefore find little support for the information-spillover hypothesis in the Chinese context; instead, a county's first IPO appears to crowd out new business formation. The effect is economically substantial: annual firm entry declines by roughly 8%, implying a meaningful contraction in local business formation when accumulated over multiple post-treatment years. The magnitude also exceeds that documented by Cornaggia et al. (2024), suggesting that IPO-induced crowding out may be particularly pronounced in China.

4.2 Event Study Results

To examine treatment dynamics and assess the plausibility of the parallel-trends assumption, Figures 2 and 3 report event-study estimates for firm entry and firm stock. The omitted period is the year immediately before a county's first IPO, and the coefficients trace the evolution of outcomes from seven years before treatment to seven years after treatment.

Figure 2 shows that the estimated coefficients are close to zero throughout the pre-treatment period and display no systematic trend, providing support for the parallel-trends assumption. Following the first IPO, the coefficients become negative and remain below zero for several years, indicating a persistent decline in new firm creation. Figure 3 exhibits a similar pattern. The

Table 2: Effect of IPOs on Firm Entry

	Log number of entering firms			
	(1)	(2)	(3)	(4)
IPO	-0.421***	-0.217***	-0.130***	-0.084**
	(0.032)	(0.028)	(0.040)	(0.033)
N	52,746	52,746	38,485	38,484
R^2	0.922	0.942	0.930	0.946
Controls × Year F.E.			✓	✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.		✓		✓

Notes: County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The dependent variable is the logarithm of the number of newly registered enterprises in the year.

pre-treatment coefficients remain statistically and economically close to zero, while the post-treatment coefficients turn negative and persist over time, suggesting a sustained contraction in the stock of active firms.

Overall, the event-study results support the empirical design in two respects. First, the absence of meaningful pre-treatment differences provides evidence consistent with the identifying assumption underlying the staggered DiD design. Second, the effects emerge only after the first IPO and remain persistent, indicating that the decline in local business formation is unlikely to reflect pre-existing trends. Together, these findings strengthen the causal interpretation that a county's first IPO reduces local business formation.

4.3 Robustness Check

This subsection conducts a series of robustness checks to assess the robustness of the baseline results. Specifically, we examine the robustness of the estimates to alternative control specifications, placebo tests, and alternative DiD estimators. Across these specifications, the main findings remain qualitatively unchanged,

Table 3: Effect of IPOs on Firm Stock

	Log stock of active firms			
	(1)	(2)	(3)	(4)
IPO	-0.450***	-0.212***	-0.113**	-0.080**
	(0.036)	(0.032)	(0.047)	(0.039)
N	52,752	52,752	38,490	38,489
R^2	0.933	0.952	0.940	0.953
Controls × Year F.E.			✓	✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.		✓		✓

Notes: County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The dependent variable is the logarithm of the number of active firms at the end of the year.

providing further support for the robustness of the baseline results.

Alternative Estimators. As a robustness check, we revisit the staggered-adoption design using the interaction-weighted estimator implemented by `eventstudyinteract`, which is designed for settings in which conventional two-way fixed-effects regressions may place problematic weights on cohort-specific treatment effects (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Sun and Abraham, 2021). To make the estimates comparable to the baseline DiD coefficients, Table 5 reports the average post-treatment effect under the counterparts of column (2) and column (4) in Tables 2 and 3. The estimates remain negative and statistically significant in all four specifications. For firm entry, the average post-treatment effect equals -0.161 in the specification with county, year, and province-by-year fixed effects and -0.073 in the full specification with additional control-by-trend interactions. For firm stock, the corresponding estimates are -0.136 and -0.078. These estimates are somewhat smaller than the baseline TWFE coefficients, but they preserve the same qualitative conclusion: after accounting for concerns about heterogeneous treatment effects in staggered DID designs, the first local IPO still

Table 4: Effect of the First Local IPO on Firm Exit

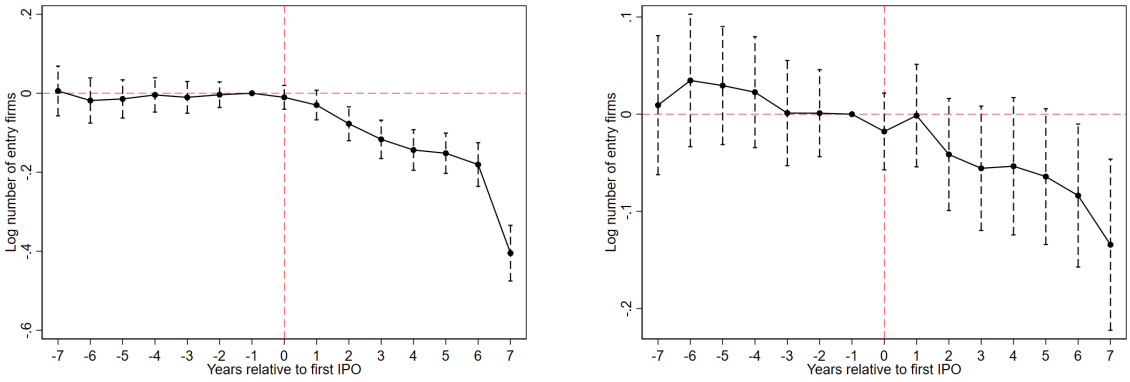
	Log number of exiting firms			
	(1)	(2)	(3)	(4)
IPO	-0.044	0.036	-0.053	0.001
	(0.047)	(0.042)	(0.060)	(0.052)
N	52,755	52,755	38,490	38,489
R^2	0.837	0.888	0.844	0.889
Controls × Year F.E.			✓	✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.		✓		✓

Notes: County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The dependent variable is the logarithm of the number of firms closed in that year.

depresses local business formation.

Figures 4 and 5 plot the corresponding event-study coefficients. The pre-treatment estimates remain close to zero overall, while the post-treatment coefficients turn negative after the first IPO year. This dynamic pattern is consistent with the baseline event-study figures and reinforces the interpretation that the contraction in business formation occurs after treatment rather than reflecting a pre-existing downward trend in treated counties.

Alternative Controls. As a first robustness check, we replace the county characteristics interacted with time fixed effects in the baseline specification with contemporaneous county-year controls. This alternative specification relaxes the functional-form assumption imposed by the baseline controls while maintaining the same set of control variables, including county-level fiscal expenditure, the number of above-scale industrial firms, GDP, population, and the value added of the secondary and tertiary sectors. Owing to data availability, the estimation sample is restricted to the period from 2000 to 2022. Table 6 shows that the main conclusions remain unchanged. In the specification with county, year, and



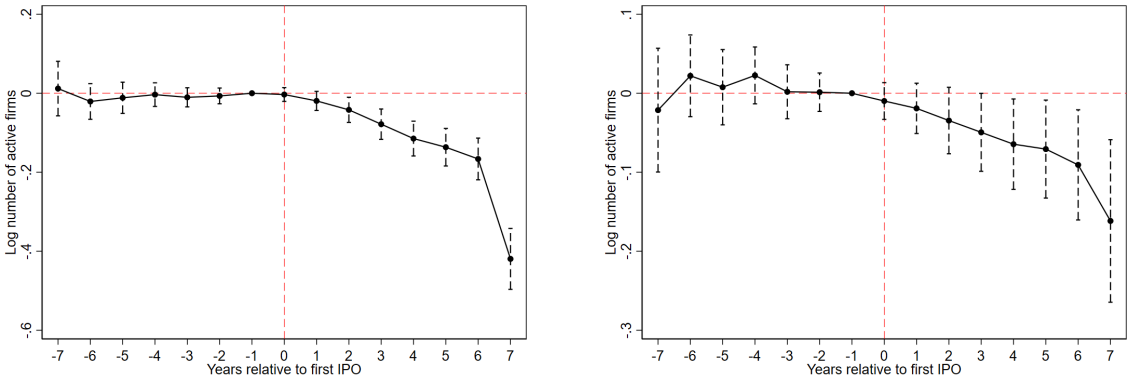
(a) Without control (b) With control

Figure 2: Dynamic Effects of the First Local IPO on Firm Entry

Notes: The figure reports the dynamic effects of the first local IPO on firm entry. Panel (a) reports estimates without controls, while Panel (b) includes controls. Vertical bars indicate 95% confidence intervals. The two panels correspond to Columns (2) and (4), respectively, of Table 2.

province-by-year fixed effects, the effects of IPO on firm entry is -0.086 and on firm stock is -0.094, both statistically significant at conventional levels, while the coefficient for firm exit remains statistically insignificant at -0.052. These findings suggest that the baseline results are robust to using contemporaneous county-year controls instead of county characteristics interacted with time fixed effects.

Cumulative Number of IPOs. A related concern is that the baseline specification focuses on the first IPO in each county and therefore does not fully exploit subsequent IPO events. To examine whether the estimated effect extends beyond the first listing, we replace the first-IPO treatment indicator with the cumulative number of IPO firms in each county and year. This specification uses all IPO events observed during the sample period and identifies the marginal effect of an additional local IPO on business formation. Table 7 reports the results. Without the additional controls, the coefficient on the cumulative number of IPOs is -0.070 for log firm entry and -0.073 for log firm stock, implying that each additional IPO is associated with roughly a 7% decline in both outcomes. These magnitudes are close to the approximately 8% effect estimated in the baseline specification for the first IPO. After including the full set of controls interacted with year fixed effects, the estimated coefficients decline in magnitude to -0.020 for entry and -0.021 for stock, but remain negative and statistically significant at conventional levels. Overall, the results indicate that the negative relationship between local IPO activity and



(a) Without control (b) With control

Figure 3: Dynamic Effects of the First Local IPO on Firm Stock

Notes: The figure reports the dynamic effects of the first local IPO on firm stock. Panel (a) reports estimates without controls, while Panel (b) includes controls. Vertical bars indicate 95% confidence intervals. The two panels correspond to Columns (2) and (4), respectively, of Table 3.

business formation is not confined to the first IPO event, although the estimated marginal effect of additional IPOs becomes more modest once differential trends associated with county characteristics are taken into account.

Excluding Counties with Multiple IPOs. One potential concern is that some counties experience multiple IPOs during the sample period, so that the estimated treatment effects may partly reflect subsequent listing events rather than the first IPO alone. In the sample, approximately 60% of IPO counties experience only one or two IPOs. As a robustness check, we therefore exclude all counties that experience more than two IPOs and re-estimate the baseline specification using only counties with one or two IPOs. Table A2 reports the results. The estimated effects remain qualitatively unchanged. The first IPO continues to reduce both new firm entry and the stock of existing firms, although the magnitudes become somewhat smaller, with the coefficient on log entry declining from -0.231 to -0.087 and that on log stock from -0.237 to -0.075. In contrast, the estimated effect on firm exit remains statistically insignificant. These findings indicate that the baseline results are not driven by counties experiencing multiple IPOs and reinforce the conclusion that the first IPO has a negative effect on local business formation.

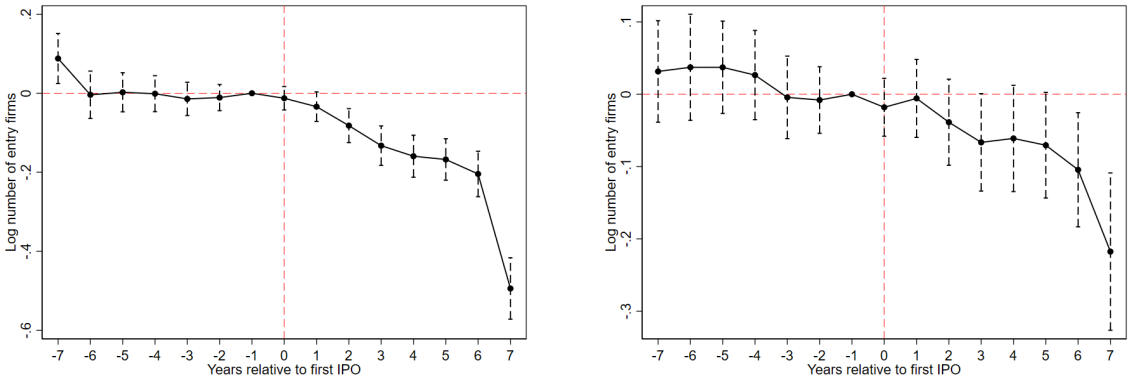
Placebo Test. As an additional robustness check, we conduct a placebo exercise based on 1,000 random assignments using the same estimation sample and the

Table 5: Robustness Check with Alternative Interaction-weighted Estimators

	Entry		Stock	
	(1)	(2)	(3)	(4)
IPO	-0.161***	-0.073***	-0.136***	-0.078***
	(0.020)	(0.028)	(0.018)	(0.025)
N	52,746	38,484	52,752	38,489
R^2	0.944	0.945	0.954	0.953
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: Columns (1) and (3) correspond to column (2) of Tables 2 and 3; columns (2) and (4) correspond to column (4) of Tables 2 and 3. The estimator implemented in the table is the interaction-weighted and heterogeneity robust DiD estimator proposed by Sun and Abraham (2021). Reported coefficients are averages of the post-treatment event-time coefficients. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

same specification as column (4) of Table 2. In each iteration, we first randomly select the same number of counties as are treated in the actual sample and assign them to the placebo treatment group. We then randomly assign placebo treatment years across these counties while preserving the empirical distribution of treatment years observed in the actual data. Figure A2 reports the resulting distribution of placebo coefficients and placebo p-values. The true estimate from the baseline specification, (-0.084), lies far in the left tail of the placebo coefficient distribution. Only 0.2% of the placebo coefficients are at least as negative as the actual estimate. At the same time, the p-values from the placebo regressions are broadly distributed as expected under random assignment, with about 10% falling below the 10% significance threshold. Taken together, these patterns indicate that the baseline result is unlikely to be generated by chance.



(a) Without control (b) With control

Figure 4: Interaction-weighted Dynamic Effects of the First Local IPO on Firm Entry

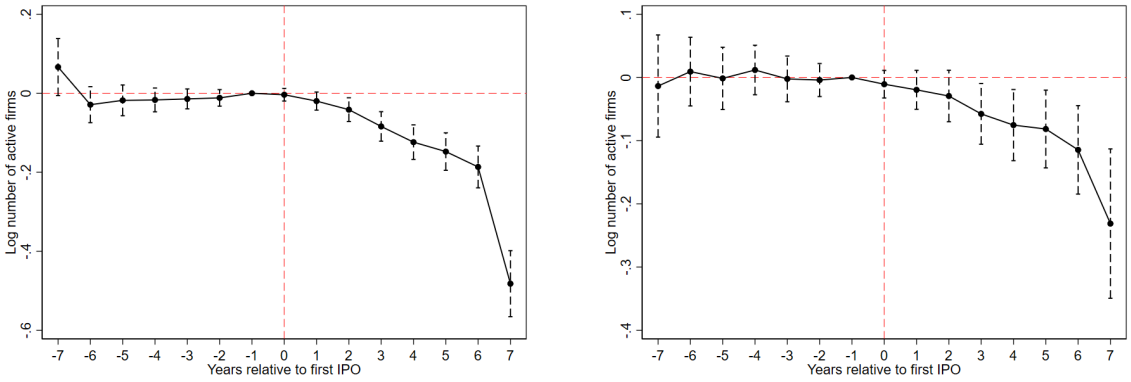
Notes: The figure reports the dynamic effects of the first local IPO on firm entry using Sun and Abraham estimators. Panel (a) reports estimates without controls, while Panel (b) includes controls. Vertical bars indicate 95% confidence intervals. The two panels correspond to Columns (1) and (2), respectively, of Table 5.

4.4 Heterogeneity

This subsection is to explore some heterogeneity regarding the negative spillover of IPOs on local business formation to gain some insights of the crowding-out channel.

Heterogeneity By Firm Type. To assess whether the decline in firm entry indeed reflects a reduction in business formation, we distinguish firms by ownership type. Private firms and self-employed businesses are the categories that most directly capture entrepreneurial entry, whereas SOEs and foreign-invested firms are less likely to represent business formation by local individuals. We therefore reclassify the original enterprise categories into three broader groups. *Private* combines self-employed businesses and private firms, *SOEs* combines state-owned enterprises, collective enterprises, and public institutions, and *Foreign* consists of foreign-invested firms. We then estimate the same regression separately using entry in each of the three groups as the dependent variable.

Panel A of Table 8 shows that the negative entry effect is strongest for the private sector and the SOE-related sector. Without the additional controls, the coefficients are -0.186 for private firms and -0.337 for SOE-type organizations, both statistically significant at the 1% level. After adding controls, the private-sector



(a) Without control (b) With control

Figure 5: Interaction-weighted Dynamic Effects of the First Local IPO on Firm Stock

Notes: The figure reports the dynamic effects of the first local IPO on firm stock using Sun and Abraham estimators. Panel (a) reports estimates without controls, while Panel (b) includes controls. Vertical bars indicate 95% confidence intervals. The two panels correspond to Columns (3) and (4), respectively, of Table 5.

coefficient remains negative and statistically significant at -0.113, while the SOE coefficient becomes smaller and insignificant. By contrast, foreign entry is not statistically different from zero in either specification. Panel B of Table 8 reports a similar but not identical pattern for the stock of active firms. The private sector again shows a robustly negative response, with coefficients of -0.226 without controls and -0.121 with controls. SOE-type firms also exhibit a large negative response in baseline specification, and the coefficient becomes statistically insignificant once controls are included. Foreign-invested firms move in the opposite direction: their stock increases significantly both without controls and with controls.

From our most stringent specification with the full set of controls, the negative effect of a local IPO is concentrated almost entirely among privately owned firms, while entry by state-owned and foreign-invested firms is largely unaffected. Moreover, the estimated effect on private firms is larger in magnitude than the corresponding effect for all firms, indicating that the decline in private business formation is the primary driver of the baseline results reported in Tables 2 and 3. This pattern suggests that the reduction in firm formation reflects a genuine crowding out of business formation, particularly among small and micro businesses established by individual entrepreneurs, rather than an

Table 6: Robustness Check with Alternative County-Year Controls

	Entry	Stock	Exit
	(1)	(2)	(3)
IPO	-0.086***	-0.094***	-0.052
	(0.031)	(0.028)	(0.046)
N	23,088	23,088	23,088
R^2	0.949	0.974	0.872
Controls	✓	✓	✓
County F.E.	✓	✓	✓
Year F.E.	✓	✓	✓
Province × Year F.E.	✓	✓	✓

Notes: This table replaces the baseline 2010 county characteristics interacted with year fixed effects by contemporaneous county-year controls. The controls include county-year fiscal expenditure, above-scale industrial firms, GDP, population, and the value added of the secondary and tertiary sectors. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

administrative response such as local governments strategically reducing the creation of state-owned firms around the time of an IPO. Taken together, these findings strengthen the interpretation that a county's first IPO discourages entrepreneurial entry by altering the incentives and opportunities faced by potential private entrepreneurs.

Heterogeneity By IPOs' ownership. Ownership structure plays an important role in China's IPO market and capital market activities, with SOEs often exhibiting substantially different behavior from non-state-owned enterprises (non-SOEs) (Chen et al., 2014; Wang, 2005). A natural question is whether the effects of IPOs on business formation differ between SOEs and non-SOEs. Table 9 examines heterogeneity by the ownership of the IPO firm. Panel A reports the estimates for counties whose IPO firm is state-owned, while Panel B reports the corresponding estimates for counties whose IPO firm is non-state-owned. The negative effects on

Table 7: Robustness Check: Effects of the Cumulative Number of IPOs

	Entry		Stock	
	(1)	(2)	(3)	(4)
IPO	-0.070***	-0.020**	-0.073***	-0.021*
	(0.007)	(0.010)	(0.008)	(0.012)
N	52,746	38,484	52,752	38,489
R^2	0.943	0.946	0.953	0.953
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: This table examines the effects of the cumulative number of IPO firms in each county on firm entry and the stock of firms. The specification provides a robustness check against potential bias arising from focusing only on a county's first IPO. The cumulative IPO measure captures the total number of firms that have gone public in a county up to each year, allowing the treatment intensity to vary with the number of local IPOs. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

business formation are substantially larger when the first IPO firm is state-owned. In the preferred specification, the coefficient on log firm entry is -0.139 for state-owned IPOs, compared with an estimate of -0.068 for non-state-owned IPOs. A similar pattern is observed for firm stock. These results suggest that the crowding-out effects of a first IPO are considerably stronger when the newly listed firm is state-owned. This finding is consistent with the institutional context of China.

Because SOEs are typically larger and employ more workers than private firms, their IPOs may generate a particularly strong demand for local labor and other productive resources. SOEs also tend to receive greater government support and subsidies and enjoy preferential access to finance, land, and other locally allocated resources. These advantages may further strengthen their ability to attract workers and compete with small and private firms in local product and factor markets.

Table 8: Effect of the First Local IPO by Firm Type

	Private		SOE		Foreign	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Entry						
IPO	-0.186***	-0.113***	-0.337***	-0.059	0.036	0.012
	(0.027)	(0.032)	(0.036)	(0.039)	(0.023)	(0.027)
N	52,755	38,489	52,755	38,489	52,755	38,489
R^2	0.952	0.955	0.650	0.640	0.802	0.768
Controls × Year F.E.		✓		✓		✓
County F.E.	✓	✓	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓	✓	✓
Panel B. Stock						
IPO	-0.226***	-0.121***	-0.212***	-0.077	0.241***	0.085**
	(0.032)	(0.039)	(0.041)	(0.054)	(0.031)	(0.038)
N	52,755	38,489	52,672	38,449	52,755	38,489
R^2	0.959	0.960	0.773	0.752	0.838	0.806
Controls × Year F.E.		✓		✓		✓
County F.E.	✓	✓	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓	✓	✓

Notes: Panel A reports the effects of the first local IPO on the entry of different types of firms, while Panel B reports the corresponding effects on the stock of active firms. *Private* combines self-employed businesses and private firms. *SOE* combines state-owned enterprises, collective enterprises, and public institutions. *Foreign* denotes foreign-invested firms. All columns include county fixed effects, year fixed effects, and province-by-year fixed effects. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Consequently, IPOs by SOEs are likely to induce greater resource reallocation toward the newly listed firm and therefore exert a stronger crowding-out effect on local business formation.

Table 9: Effect of the First Local IPO by Ownership of the IPO Firm

	Entry		Stock	
	(1)	(2)	(3)	(4)
Panel A. State-owned IPO firms				
IPO	-0.310***	-0.139**	-0.308***	-0.136**
	(0.042)	(0.056)	(0.049)	(0.069)
N	43,794	32,468	43,799	32,473
R^2	0.941	0.944	0.950	0.950
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓
Panel B. Non-state-owned IPO firms				
IPO	-0.167***	-0.068*	-0.161***	-0.054
	(0.034)	(0.038)	(0.037)	(0.042)
N	43,738	34,360	43,744	34,365
R^2	0.941	0.944	0.950	0.951
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: Treated counties are divided according to whether the first local IPO firm is SOE or non-SOE, while the control counties are common across the two panels. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Regional Heterogeneity. China's regional development has long been characterized by substantial disparities, with the eastern region enjoying higher levels of economic development, deeper financial markets, and a more diversified

industrial structure, while the central and western regions remain relatively less developed. As a result, firms in less-developed regions are likely to face a more constrained local resource environment. If the crowding-out effect of a local IPO operates through the reallocation of scarce local resources, it should be more pronounced in regions where such resources are relatively scarce.

Table 10 reports heterogeneous effects across China's eastern, central, and western regions. The results are consistent with this prediction. In the preferred specification, the estimated effect on firm entry is -0.115 in the eastern region, compared with -0.295 and -0.417 in the central and western regions, respectively. A similar regional gradient is observed for the stock of active firms. Overall, these findings suggest that the crowding-out effect of a local IPO becomes stronger as local regions become less developed, consistent with the view that the IPO firm reallocates scarce entrepreneurial resources away from existing and potential local firms.

Table 10: Heterogeneous Effects by Region

	Entry			Stock		
	(1)	(2)	(3)	(4)	(5)	(6)
	East	Central	West	East	Central	West
IPO	-0.115***	-0.295***	-0.417***	-0.100***	-0.288***	-0.459***
	(0.035)	(0.053)	(0.070)	(0.038)	(0.061)	(0.081)
N	44,057	40,228	36,457	44,062	40,233	36,463
R^2	0.944	0.937	0.938	0.954	0.946	0.947
County F.E.	✓	✓	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓	✓	✓

Notes: Columns (1)–(3) report the effects of the first local IPO on firm entry for the eastern, central, and western regions, respectively. Columns (4)–(6) report the corresponding effects on the stock of active firms. All specifications include county fixed effects, year fixed effects, province-by-year fixed effects, and the baseline control-by-year-FE interactions. Standard errors clustered at the county level are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

5 Mechanism

To better understand the channels through which IPOs crowd out local business formation and firm entry, this section explores the underlying mechanisms and examines the empirical predictions derived from Hypothesis 2.

5.1 Industry Competition Effects

One potential mechanism behind the crowding-out effect is intensified competition generated by the IPO firm. Following the IPO, the newly listed firm may strengthen its competitive position through greater access to financing, labor, land, government support, and other business opportunities. Its expansion may therefore place competing local firms under greater pressure in both product and factor markets. Because firms in the same industry compete more directly for these resources and market opportunities, this mechanism predicts that the crowding-out effect should be more pronounced among firms operating in the IPO firm's industry (Cornaggia et al., 2024).

To examine this mechanism, we classify firms according to whether they operate in the same broad industry as the IPO firm, based on the one-letter CSRC industry classification, and construct separate measures of firm entry and firm stock for the two groups. We then re-estimate the baseline specification using each set of outcome variables following. Table 11 reports the estimated effects separately for firms operating in the same and different industries as the IPO firm. The results reveal a striking contrast between the effects of IPOs on the two groups. For firms in the same industry, the first local IPO reduces firm entry by 92.2% in the baseline specification and by 68.5% after including the controls. The corresponding effects on the stock of active firms are also large, at 67.1% and 40.8%, respectively, and remain highly statistically significant across specifications.

By contrast, the estimated effects on firms in different industries are much smaller. For firm entry, the coefficient falls from -0.922 in the baseline specification to -0.119, and further to 0.011 and statistically insignificant after adding the full set of controls. A similar pattern emerges for firm stock: the coefficient declines from -0.671 to -0.147 and then to -0.020, which is also statistically insignificant in the most stringent specification. Figure A3 reports the corresponding event-study

estimates for firms in the same and different industries and confirms this pattern. Overall, the evidence indicates that the crowding-out effect of a local IPO is concentrated among firms that compete more directly with the IPO firm. This finding is consistent with Cornaggia et al. (2024), who show that the negative local effects of IPOs are driven primarily by sector peers. It also supports the view that intensified competition for industry-specific resources—including labor, customers, and other production inputs—plays an important role in suppressing local business formation following a county's first IPO.

5.2 Labor Market Reallocation

The preceding results show that the crowding-out effect of a local IPO on business formation is concentrated among firms operating in the same industry as the IPO firm. However, the industry-level evidence alone cannot distinguish whether this pattern is driven by competition in factor markets, product markets, or both. One plausible explanation is that the newly listed firm competes more aggressively for labor following the IPO, attracting workers away from incumbent firms and potential entrepreneurial ventures. Following an IPO, the newly listed firm typically expands its operations and increases labor demand (Babina et al., 2025; Borisov et al., 2021). As one of the largest and most visible employers in the county, the listed firm may attract workers who would otherwise be employed by existing firms or start new businesses. If entrepreneurial talent and skilled labor are locally scarce, this reallocation can reduce the supply of potential entrepreneurs, thereby lowering new firm entry and slowing business formation. This hypothesis yields a straightforward empirical prediction: if labor market reallocation is an important mechanism, the negative effect of a local IPO on business formation should be stronger when the newly listed firm employs a larger workforce. To test this prediction, we interact the IPO treatment indicator with the employment size of the IPO firm. A significantly more negative interaction coefficient would be consistent with the view that labor reallocation toward the listed firm contributes to the decline in local business formation.

Table 12 examines whether the labor reallocation mechanism is stronger when the IPO firm employs more workers by interacting the IPO treatment indicator with the employment size of the IPO firm. The interaction term is negative

Table 11: Effect of the First Local IPO by Industry Similarity

	Same Industry		Different Industry	
	(1)	(2)	(3)	(4)
Panel A. Firm Entry				
IPO	-0.922***	-0.685***	-0.119***	0.011
	(0.046)	(0.053)	(0.029)	(0.035)
N	52,686	38,458	52,746	38,484
R^2	0.886	0.910	0.941	0.944
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓
Panel B. Firm Stock				
IPO	-0.671***	-0.408***	-0.147***	-0.020
	(0.048)	(0.054)	(0.033)	(0.040)
N	52,751	38,488	52,752	38,489
R^2	0.905	0.922	0.951	0.951
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: Panel A reports the estimated effects of the first local IPO on firm entry, while Panel B reports the effects on the stock of active firms. Within each panel, columns (1) and (2) restrict the outcome to firms operating in the same industry as the IPO firm, whereas columns (3) and (4) restrict the outcome to firms in different industries. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

and statistically significant across all specifications for both firm entry and firm stock. The estimates indicate that counties whose first IPO firm employs a larger workforce experience a greater decline in entrepreneurial entry and the stock of active firms following the IPO. These findings are consistent with the labor reallocation hypothesis, suggesting that larger IPO firms may absorb more local labor and thereby exert a stronger crowding-out effect on business formation. Figure A4 provides further evidence by splitting IPO firms into high- and low-employment groups according to the median employment size and estimating the event-study specification separately for the two groups. The event-study estimates show that the decline in business formation is substantially larger following IPOs of high-employment firms, while the corresponding effects for low-employment firms are considerably weaker. Together, the interaction analysis and the event-study evidence provide complementary support for the view that labor market reallocation is an important channel through which local IPOs reduce business formation.

One potential concern is that the employment size of the IPO firm may simply proxy for overall firm size. Larger firms may crowd out local business formation not only by hiring more workers, but also by attracting other scarce local resources, such as financial capital and government support. To distinguish the labor reallocation mechanism from a more general firm-size effect, we replace employment with the IPO firm's registered capital and estimate the same interaction specification. Table A3 reports the results using registered capital. The interaction term is statistically insignificant across all specifications for both firm entry and firm stock, providing little evidence that the crowding-out effect systematically varies with the IPO firm's registered capital. Together with the employment results, these findings suggest that the stronger crowding-out effect associated with larger employers is unlikely to be driven solely by overall firm size, and is instead more consistent with the labor reallocation mechanism.

This interpretation is also consistent with existing evidence on China's IPO selection process. Johansson et al. (2017) show that firms with greater excess employment are more likely to obtain IPO approval, particularly among private firms, suggesting that employment considerations play an important role in the IPO approval process. Our findings also complement Fang et al. (2023):

whereas the closure of state-owned enterprises pushed workers into reluctant entrepreneurship, local IPOs create employment opportunities and crowd out reluctant entrepreneurship. Combined with the findings in this paper, the evidence suggests that the labor market may be an important channel through which IPOs affect local business formation. In particular, IPO firms with larger workforces are likely to create more employment opportunities after going public, increasing the attractiveness of paid employment relative to entrepreneurship. As a result, potential entrepreneurs may choose wage employment at the newly listed firm instead of starting their own businesses, leading to a larger decline in both entrepreneurial entry and the stock of local firms.

Table 12: Heterogeneous Effects by IPO Employment Size

	Entry		Stock	
	(1)	(2)	(3)	(4)
IPO × Log Employment	-0.065***	-0.053*	-0.064**	-0.068**
	(0.025)	(0.031)	(0.026)	(0.034)
IPO	0.258	0.301	0.255	0.415*
	(0.181)	(0.222)	(0.191)	(0.241)
N	52,746	38,484	52,752	38,489
R^2	0.943	0.946	0.952	0.953
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: This table examines whether the effect of a county's first IPO varies with firm employment size. The interaction term is constructed by interacting the IPO indicator with firms' employment size. Columns (1) and (3) report the baseline specification, while columns (2) and (4) additionally include county characteristics interacted with year fixed effects. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

5.3 Financial Accessibility

A large body of literature has documented the importance of financial development and access to external finance for entrepreneurship, showing that financing constraints can substantially hinder business formation and firm growth (Bianchi, 2012; King and Levine, 1993; Love and Martínez Pería, 2015). Against this background, another potential mechanism is financial accessibility. Following an IPO, the newly listed firm may obtain easier access to bank credit and government-supported financing, thereby strengthening its ability to attract local financial resources. If local credit markets are imperfect, the expansion of the IPO firm may reduce the availability of financing for incumbent firms and potential entrepreneurs, discouraging new business formation and constraining the growth of existing firms.

This mechanism yields a straightforward empirical prediction. If competition for financial resources is an important channel, the crowding-out effect of a local IPO should be stronger in counties with more limited access to finance, where local firms face tighter financing constraints and the reallocation of credit toward the newly listed firm is likely to have a larger impact. Following Peng et al. (2025), we measure local financial accessibility by the number of financial institution branches per capita in each county. We then test this prediction by interacting the IPO treatment indicator with this measure of financial accessibility.

Table 13 reports the heterogeneous effects by local financial institution density. If competition for financial resources were an important mechanism, the crowding-out effect of a local IPO would be expected to be stronger in counties with fewer financial institution branches. However, the results provide little support for this hypothesis. In the baseline specification, the interaction between the IPO indicator and the number of financial institution branches per capita is negative and statistically significant for both firm entry and firm stock, suggesting that the negative effects of a local IPO are, if anything, larger in counties with greater financial institution density. Moreover, once county characteristics interacted with time fixed effects are included, the interaction term becomes statistically insignificant for both outcomes. Overall, these findings do not support the view that competition for local financial resources is the primary mechanism through which a local IPO crowds out business formation.

Table 13: Heterogeneous Effects by Local Financial Institution Density

	Entry		Stock	
	(1)	(2)	(3)	(4)
IPO × Financial Branches per Capita	-0.014***	-0.001	-0.007*	0.006
	(0.004)	(0.006)	(0.004)	(0.006)
IPO	-0.102**	-0.078	-0.155***	-0.119**
	(0.042)	(0.054)	(0.046)	(0.058)
N	46,107	38,451	46,110	38,456
R^2	0.943	0.946	0.952	0.953
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: This table examines whether the effect of a county's first IPO varies with local financial institution density. The interaction term is constructed by interacting the IPO indicator with the number of financial institution branches per capita in the county. Columns (1) and (3) report the baseline specification, while columns (2) and (4) additionally include county characteristics interacted with time fixed effects. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

6 Conclusion

This paper examines how local IPO activity affects business formation in China. Using county-level panel data from 1990 to 2022 and a staggered difference-in-differences design, we find that business formation declines after a county experiences an IPO. Our preferred estimates indicate that the presence of a local IPO is associated with declines of approximately 8% in both new firm entry and the stock of active firms, while having little effect on firm exit. Event-study estimates show no evidence of differential pre-trends and indicate that the negative effects emerge only after the IPO.

The crowding-out effect is not uniform across firms or IPOs. It is strongest for private-sector business formation, weaker for state-owned enterprises, and insignificant for foreign-invested firms. Moreover, the effect is substantially stronger when the newly listed firm is state-owned, whereas IPOs by

non-state-owned firms generate much smaller or statistically insignificant effects. These findings suggest that government involvement plays an important role in shaping how the benefits associated with public listing are redistributed within the local economy.

The mechanism analysis provides evidence that the crowding-out effect is driven primarily by intensified competition within industries and the reallocation of labor rather than by competition for financial resources. The negative effect is concentrated among firms operating in the same industry as the IPO firm, consistent with intensified competition for industry-specific resources. Additional evidence suggests that labor market reallocation is an important transmission channel, while there is little evidence that competition for local financial resources explains the observed decline in business formation.

Overall, this paper provides new evidence from China on how IPOs affect local business formation. The findings suggest that a local IPO should not be viewed as an unambiguously positive shock to the local entrepreneurial ecosystem. Instead, in the Chinese institutional context, public listing appears to reallocate scarce local resources toward the newly listed firm, with the benefits accruing disproportionately to listed firms while imposing substantial negative spillovers on private business formation. More broadly, the paper highlights the importance of considering resource reallocation, ownership structure, and local institutional arrangements when evaluating the regional consequences of capital market development.

7 Disclosure of Interest

All authors disclosed no relevant relationships or interests.

8 Funding

There is no funding associated with the article.

9 Data Availability Statement

Data will be made available on request.

10 Declaration of Generative AI

Generative AI was used only in the writing process to improve readability and language.

References

Akcigit, U., Baslandze, S., and Lotti, F. (2023). Connecting to power: Political connections, innovation, and firm dynamics. *Econometrica*, 91(2):529–564.

Almeida, H., Cunha, I., Ferreira, M. A., and Restrepo, F. (2017). The real effects of credit ratings: The sovereign ceiling channel. *Journal of Finance*, 72(1):249–290.

Babina, T., Ouimet, P., and Zarutskie, R. (2025). Ipos, human capital, and labor reallocation. *Journal of Financial and Quantitative Analysis*, 60(6):2584–2614.

Barrios, J. M., Choi, J. H., Hochberg, Y. V., Kim, J., and Liu, M. (2026). Initial public offering and new business formation: The role of public firm disclosures. *Journal of Accounting Research*. Forthcoming.

Barwick, P. J., Chen, L., Li, S., and Zhang, X. (2025). Entry deregulation, market turnover, and efficiency: China's business registration reform. *The Review of Economics and Statistics*, pages 1–46.

Beck, T. and Levine, R. (2004). Stock markets, banks, and growth: Panel evidence. *Journal of Banking & Finance*, 28(3):423–442.

Bekaert, G., Harvey, C. R., and Lundblad, C. (2005). Does financial liberalization spur growth? *Journal of Financial Economics*, 77(1):3–55.

Bernstein, S. (2015). Does going public affect innovation? *Journal of Finance*, 70(4):1365–1403.

Bernstein, S., Colonnelli, E., Giroud, X., and Iverson, B. (2019). Bankruptcy spillovers. *Journal of Financial Economics*, 133(3):608–633. JFE Special Issue on Labor and Finance.

Bianchi, M. (2012). Financial development, entrepreneurship, and job satisfaction. *Review of Economics and Statistics*, 94(1):273–286.

BIERMANN, M. and HUBER, K. (2024). Tracing the international transmission of a crisis through multinational firms. *The Journal of Finance*, 79(3):1789–1829.

Black, S. E. and Strahan, P. E. (2002). Entrepreneurship and bank credit availability. *Journal of Finance*, 57(6):2807–2833.

Borisov, A., Ellul, A., and Sevilir, M. (2021). Access to public capital markets and employment growth. *Journal of Financial Economics*, 141(3):896–918.

Branstetter, L. G., Li, G., and Ren, M. (2023). Picking winners? government subsidies and firm productivity in china. *Journal of Comparative Economics*, 51(4):1186–1199.

Butler, A. W., Fauver, L., and Spyridopoulos, I. (2019). Local economic spillover effects of stock market listings. *Journal of Financial and Quantitative Analysis*, 54(3):1025–1050.

Callaway, B. and Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230.

Cetorelli, N. and Strahan, P. E. (2006). Finance as a barrier to entry: Bank competition and industry structure in local u.s. markets. *Journal of Finance*, 61(1):437–461.

Chen, C., Shi, H., and Xu, H. (2014). The ipo underwriting market share in china: Do ownership and quality matter? *Journal of Banking & Finance*, 46:177–189.

Chen, J., Heng, C. S., Tan, B. C., and Lin, Z. (2018). The distinct signaling effects of r&d subsidy and non-r&d subsidy on ipo performance of it entrepreneurial firms in china. *Research policy*, 47(1):108–120.

Chen, S., Mao, H., and Feng, Z. (2020). Political uncertainty and firm entry: Evidence from chinese manufacturing industries. *Journal of Business Research*, 120:16–30.

Cong, L. W. and Howell, S. T. (2021). Policy uncertainty and innovation: Evidence from initial public offering interventions in china. *Management Science*, 67(11):7238–7261.

Cornaggia, J., Gustafson, M., Kotter, J., and Pisciotta, K. (2024). Initial public offerings and the local economy: Evidence of crowding out. *Review of Finance*, 28(5):1245–1273.

Da Rin, M., Di Giacomo, M., and Sembenelli, A. (2011). Entrepreneurship, firm entry, and the taxation of corporate income: Evidence from europe. *Journal of public economics*, 95(9-10):1048–1066.

Demirci, I., Huang, J., and Sialm, C. (2019). Government debt and corporate leverage: International evidence. *Journal of Financial Economics*, 133(2):337–356.

Fang, H., Li, M., Wu, Z., and Zhang, Y. (2023). Reluctant entrepreneurs: Evidence from china's soe reform. Working Paper 31700, National Bureau of Economic Research.

Gompers, P., Lerner, J., and Scharfstein, D. (2005). Entrepreneurial spawning: Public corporations and the genesis of new ventures, 1986 to 1999. *Journal of Finance*, 60(2):577–614.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

Hartman-Glaser, B., Thibodeau, M., and Yoshida, J. (2023). Cash to spend: Ipo wealth and house prices. *Real Estate Economics*, 51(1):68–102.

Hopenhayn, H. A. (1992). Entry, exit, and firm dynamics in long run equilibrium. *Econometrica*, 60(5):1127–1150.

HSU, H.-C., REED, A. V., and ROCHOLL, J. (2010). The new game in town: Competitive effects of ipos. *The Journal of Finance*, 65(2):495–528.

Huang, Y., Pagano, M., and Panizza, U. (2020). Local crowding-out in china. *Journal of Finance*, 75(6):2855–2898.

Jiang, F., Lowry, M., and Qian, Y. (2024). Local ipos and household stock market participation. *Review of Finance*, 28(6):1919–1952.

Johansson, A. C., Luo, D., Rickne, J., and Zheng, W. (2017). Government intervention in the capital allocation process: Excess employment as an ipo selection rule in china. *China Economic Review*, 44:271–281.

Kaplan, D. S., Piedra, E., and Seira, E. (2011). Entry regulation and business start-ups: Evidence from mexico. *Journal of Public Economics*, 95(11-12):1501–1515.

Kerr, W. R. and Nanda, R. (2009). Democratizing entry: Banking deregulations, financing constraints, and entrepreneurship. *Journal of Financial Economics*, 94(1):124–149.

King, R. G. and Levine, R. (1993). Finance, entrepreneurship and growth. *Journal of Monetary economics*, 32(3):513–542.

Klapper, L., Laeven, L., and Rajan, R. (2006). Entry regulation as a barrier to entrepreneurship. *Journal of financial economics*, 82(3):591–629.

Li, X., Zhang, B., and Yu, K. (2026). Does improved ipo access stimulate venture capital? evidence from china's registration-based reform. *Pacific-Basin Finance Journal*, 99:103235.

Lin, H., Shentu, L., and You, J. (2026). Ipo boom, neighbour's gloom: The hidden costs of ipos on local private smes. *European Financial Management*, 32(4):1089–1119.

Love, I. and Martínez Pería, M. S. (2015). How bank competition affects firms' access to finance. *The World Bank Economic Review*, 29(3):413–448.

Maksimovic, V., Phillips, G., and Yang, L. (2023). Do ipo firms become myopic? *Review of Finance*, 27(3):765–807.

Martin, T. (2025). Real effects of centralized markets: Evidence from steel futures. *The Review of Financial Studies*, 38(7):2140–2181.

Matray, A. (2021). The local innovation spillovers of listed firms. *Journal of Financial Economics*, 141(2):395–412.

Michelacci, C. and Silva, O. (2007). Why so many local entrepreneurs? *Review of Economics and Statistics*, 89(4):615–633.

Peng, Y., Yang, J., Shen, J., and Gou, Q. (2025). Financial outreach, bank deposits, and economic growth. *Journal of Economic Dynamics and Control*, 171:105036.

Qian, Y., Ritter, J. R., and Shao, X. (2024). Initial public offerings chinese style. *Journal of Financial and Quantitative Analysis*, 59(1):1–38.

Samila, S. and Sorenson, O. (2011). Venture capital, entrepreneurship, and economic growth. *Review of Economics and Statistics*, 93(1):338–349.

Spiegel, M. and Tookes, H. (2020). Why does an ipo affect rival firms? *The Review of Financial Studies*, 33(7):3205–3249.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199.

Wang, C. (2005). Ownership and operating performance of chinese ipos. *Journal of banking & finance*, 29(7):1835–1856.

Online Appendix

A Addtional Tables

Table A1: Summary Statistics of County Characteristics

	All Counties		Treated Counties		Control Counties	
Variable	Obs	Mean	Obs	Mean	Obs	Mean
Log public expenditure	1,423	11.57	439	11.94	984	11.40
Log number of above-scale industrial firms	1,207	4.15	340	5.29	867	3.70
Secondary-industry share	1,363	0.45	403	0.52	960	0.41
Tertiary-industry share	1,364	0.35	404	0.36	960	0.34
Log GDP	1,401	13.41	431	14.33	970	13.01
Log population	1,430	3.55	455	3.95	975	3.36

Notes: This table reports summary statistics for county characteristics measured in 2010. Treated counties are counties that experience at least one IPO during the sample period, while control counties are counties that never experience an IPO.

Table A2: Robustness Check: Excluding Counties with Multiple IPOs

	Entry		Stock		Exit	
	(1)	(2)	(3)	(4)	(5)	(6)
IPO	-0.231***	-0.087**	-0.237***	-0.075	-0.021	-0.008
	(0.036)	(0.040)	(0.041)	(0.047)	(0.055)	(0.064)
N	44,926	34,855	44,932	34,860	44,935	34,860
R^2	0.938	0.942	0.947	0.949	0.881	0.884
Controls × Year F.E.		✓		✓		✓
County F.E.	✓	✓	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓	✓	✓

Notes: This table use the sample excluding counties that experience multiple IPOs during the sample period. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A3: Heterogeneous Effects by IPO Registered Capital

	Entry		Stock	
	(1)	(2)	(3)	(4)
IPO × Log capital	-0.055*	-0.050	-0.050	-0.041
	(0.030)	(0.033)	(0.032)	(0.035)
IPO	0.835	0.858	0.739	0.685
	(0.565)	(0.634)	(0.613)	(0.665)
N	52,746	38,484	52,752	38,489
R^2	0.943	0.945	0.952	0.952
Controls × Year F.E.		✓		✓
County F.E.	✓	✓	✓	✓
Year F.E.	✓	✓	✓	✓
Province × Year F.E.	✓	✓	✓	✓

Notes: This table examines whether the effect of a county's first IPO varies with IPO registered capital, measured by the logarithm of the registered capital. Columns (1) and (3) report the baseline specification, while columns (2) and (4) additionally include county characteristics interacted with time fixed effects. County-clustered standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B Addtional Figures

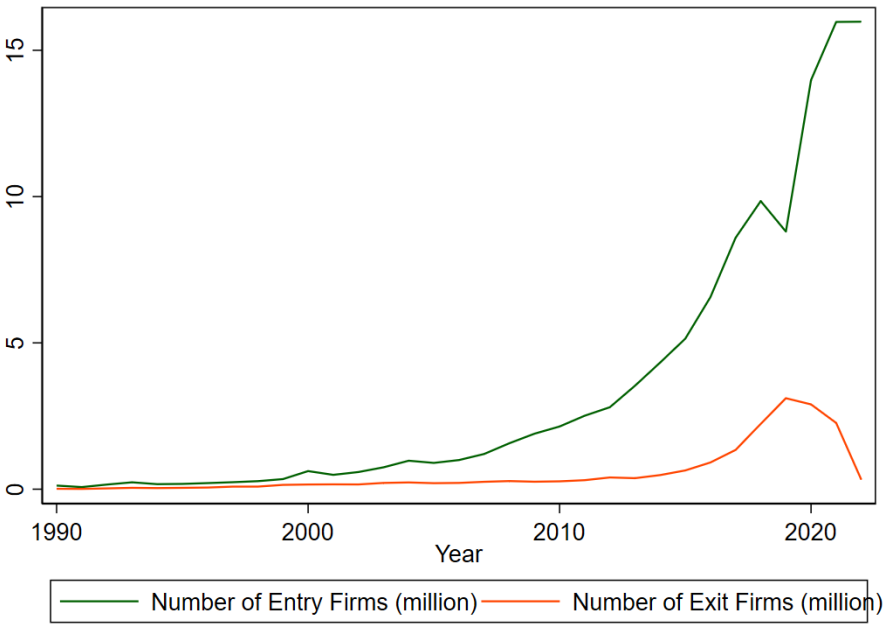
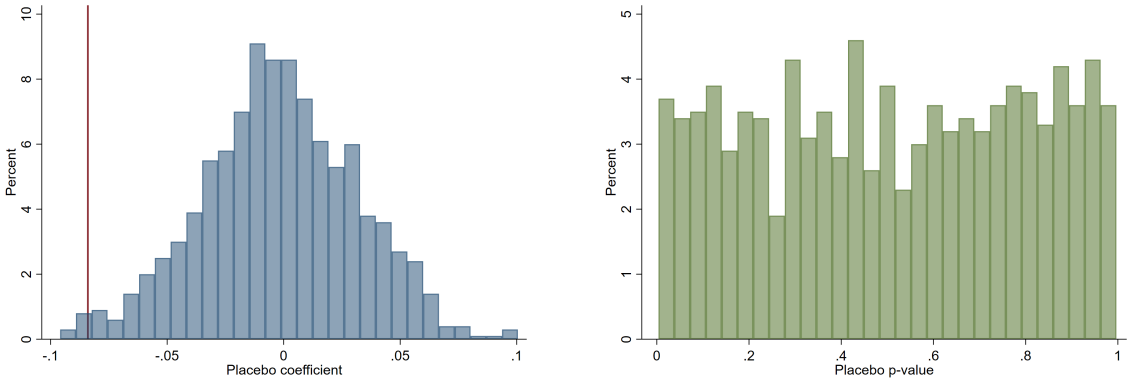


Figure A1: Number of Entry and Exit Firms Over Time

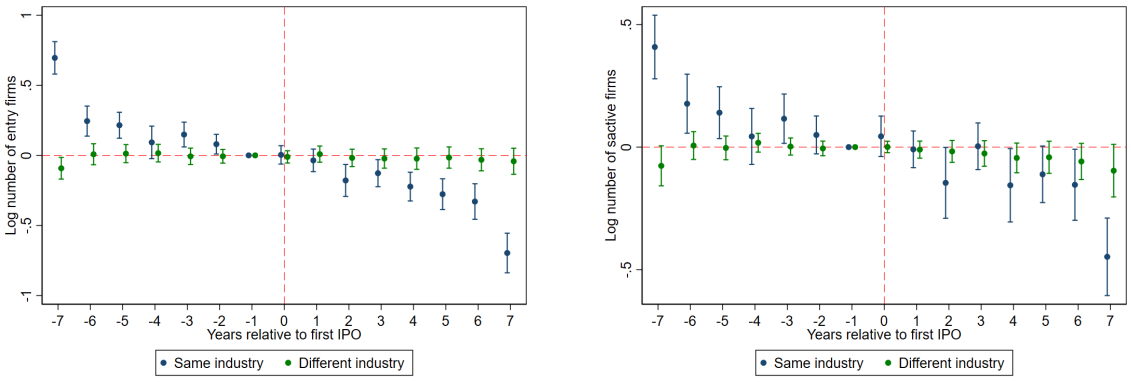
Notes: The figure plots the evolution of firm entry and firm exit in China from 1990 to 2022. Firm-level information is obtained from China's National Enterprise Registration System, the official nationwide business registration and disclosure system.



(a) Coefficient Distribution (b) P-value Distribution

Figure A2: Placebo Test Based on 1,000 Random Treatment Assignments

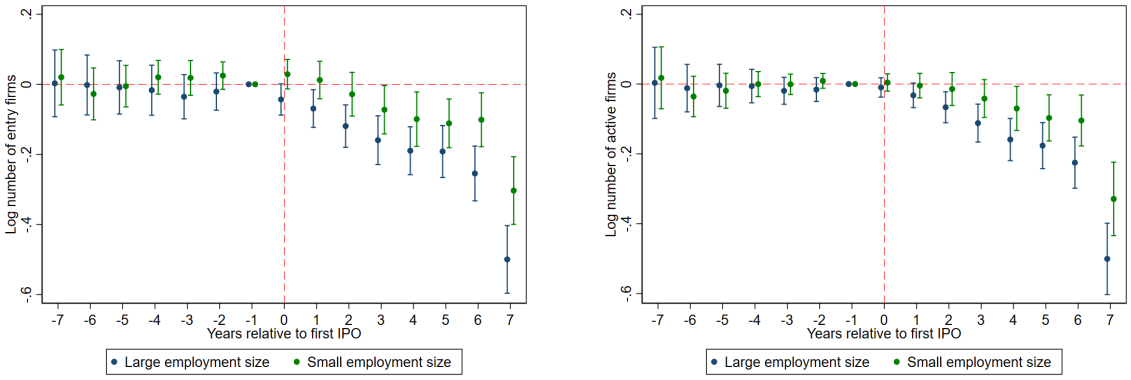
Notes: This figure reports the results of 1,000 placebo treatment assignments using the same estimation sample and specification as column (4) of Table 2. In each iteration, we randomly select the same number of counties as are treated in the actual sample and then randomly assign placebo treatment years to these counties while preserving the empirical distribution of the actual first-IPO years. Panel (a) shows the distribution of the placebo coefficients, and Panel (b) shows the corresponding distribution of placebo p-values. The vertical line in Panel (a) indicates the baseline estimate of -0.084.



(a) Firm Entry (b) Firm Stock

Figure A3: Event Study: Same Industry vs. Different Industries

Notes: This figure presents event-study estimates of the effects of local IPOs on firms in different industry groups. Panel (a) reports the effects on firm entry, while Panel (b) reports the effects on the stock of firms. In each panel, one set of estimates compares firms in the same industry as the local IPO firm with firms in control counties, while the other compares firms in industries different from that of the local IPO firm with firms in control counties.



(a) Firm Entry

(b) Firm Stock

Figure A4: Event Study: Small Employment vs. Large Employment

Notes: This figure presents event-study estimates separately for counties experiencing IPOs by firms with relatively small and large employment. IPO firms are classified as small or large based on whether their employment is below or above the median employment of IPO firms in the sample. Panel (a) reports the effects on firm entry, while Panel (b) reports the effects on the stock of firms. In each panel, the estimates compare treated counties in the small- and large-employment groups, respectively, with control counties that do not experience an IPO during the sample period.